

Deep Learning-Based Athlete Injury Risk Analyzer with Explainable AI

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Abstract— Sports-related injuries impair athletic performance and long-term musculoskeletal health, and their multifactorial nature makes rule-based screening unreliable. This paper presents a deep-learning decision-support prototype that estimates the probability of elevated injury risk from fifteen demographic, anthropometric, training-load, recovery, and self-reported variables. The core predictive model is a fully connected deep neural network (multi-layer perceptron, MLP) with a 128–64–32–16 hidden-unit topology, ReLU activations, batch normalization, and dropout regularization, trained with the Adam optimizer under early stopping and adaptive learning-rate scheduling. A complementary symmetric autoencoder is independently trained to learn a sixteen-dimensional unsupervised latent representation of the same standardized feature space. The supervised MLP was trained on 800 records and evaluated once on a stratified, held-out test set of 200 records. At the default decision threshold of 0.50, the network achieved an accuracy of 0.8850, precision of 0.6923, recall of 0.7105, F1-score of 0.7013, and ROC-AUC of 0.9108, exceeding a logistic-regression baseline on every reported metric. To support transparent decision-making, the deployed system integrates SHAP and LIME to generate per-athlete, locally faithful explanations alongside a three-band risk label and rule-based training recommendations, delivered through an interactive Streamlit interface. The reported results are encouraging for a coursework-level proof of concept; however, they do not constitute clinical validation. Dataset provenance, label construction, external generalizability, probability calibration, and the operating decision threshold all require prospective, multi-site evaluation before the system could be considered for return-to-play or medical decision support.

Index Terms— athlete injury risk prediction, deep learning, multi-layer perceptron, autoencoder, explainable artificial intelligence, SHAP, LIME, tabular data classification, sports informatics.

I. INTRODUCTION

Sports participation confers well-established physical and psychosocial benefits, yet it exposes athletes to a persistent risk of musculoskeletal injury that can curtail performance, prolong rehabilitation, and, in severe cases, end competitive careers. Injury occurrence is widely recognized as multifactorial: prior injury history, cumulative training exposure, recovery quality, anthropometric profile, sleep behavior, psychological stress, and sport-specific mechanical loading interact through pathways that are frequently non-linear and athlete-specific [1]–[6]. Because no single univariate threshold captures this complexity, screening tools built on fixed cut-offs or simple heuristics tend to generalize poorly across athletes and disciplines.

Data-driven modeling offers a complementary approach: rather than encoding an a priori causal rule, a learned model can

summarize statistical regularities present in routinely collected athlete-monitoring variables. Deep neural networks are particularly well suited to tabular health and performance data because they can approximate non-linear interactions among heterogeneous predictors without manual feature engineering [8]–[10]. However, a model intended to support—rather than replace—coaching and clinical judgment must go beyond raw predictive accuracy. It must (i) quantify predictive uncertainty, (ii) avoid implicit causal claims that the underlying observational data cannot support, and (iii) remain interpretable to non-specialist stakeholders such as coaches, athletes, and support staff [7]–[10].

This paper describes the design, implementation, and evaluation of a reproducible proof-of-concept system for binary athlete injury-risk classification from fifteen tabular predictors. The contribution is organized around four deliverables: (1) a leakage-aware preprocessing pipeline with a stratified train/test partition; (2) a regularized multi-layer perceptron classifier trained with modern deep-learning regularization and optimization practice; (3) an unsupervised symmetric autoencoder that independently learns a compact latent representation of the same feature space; and (4) an interactive Streamlit application that surfaces risk probabilities alongside local SHAP and LIME explanations and rule-based recommendations. Throughout, the system output is explicitly framed as a risk estimate for a coursework prototype—not a diagnosis, not a causal explanation, and not a substitute for a qualified sports-medicine professional.

II. LITERATURE REVIEW

Foundational sports-medicine literature frames injury as the outcome of a dynamic, multifactorial process rather than a single dominant risk factor. Meeuwisse's multifactorial model [2] and its complex-systems extension by Bittencourt et al. [3] argue that injury emerges from the interaction of internal risk factors (e.g., prior injury, anthropometry) and external exposures (e.g., training load), mediated by an inciting event. Workload-based epidemiology has further shown that both under-training and abrupt spikes in training load are associated with elevated injury risk, motivating the inclusion of training-frequency, duration, and intensity variables in predictive models [4]–[6].

A parallel body of work has applied supervised machine learning to injury-risk estimation. Systematic reviews by Claudino et al. [8] and Van Eetvelde et al. [9] catalog a growing set of studies applying logistic regression, tree ensembles, support vector machines, and neural networks to athlete-monitoring data, generally reporting moderate-to-good discrimination but highlighting recurring methodological weaknesses: small sample sizes, inconsistent outcome definitions, and limited external validation. Bullock et al. [10] specifically caution that many published injury-prediction

models exhibit optimistic performance estimates attributable to single-split evaluation and a lack of calibration reporting — concerns that directly motivate the cautious, non-clinical framing adopted in this project.

On the technical side, the regularization and optimization techniques used here are drawn from well-established deep-learning literature: rectified linear activation [14], batch normalization [15], dropout [16], and the Adam optimizer [18], together with early-stopping heuristics for controlling overfitting [19]. Autoencoder-based unsupervised representation learning traces to Hinton and Salakhutdinov [21] and has since been extended to denoising and general-purpose feature-extraction variants [22], [23]. For model interpretability, SHAP [24], [25] and LIME [26] are the two dominant post-hoc, model-agnostic explanation frameworks used in this work; both carry theoretical caveats regarding stability and susceptibility to adversarial manipulation [27]–[29], which this paper explicitly discusses rather than glosses over.

III. PROBLEM STATEMENT

Given a vector of fifteen standardized athlete attributes spanning demographic, anthropometric, training-exposure, recovery, and injury-history domains, the objective is to learn a function that estimates the probability that an athlete belongs to the high-injury-risk class, as opposed to the low-injury-risk class, under a binary label $\text{Injury_Risk} \in \{0, 1\}$. The problem is formulated as supervised binary classification on tabular, cross-sectional (non-longitudinal) data. Three constraints shape the solution: (i) the dataset exhibits moderate class imbalance (19.1% positive class), so accuracy alone is an insufficient evaluation criterion; (ii) the label-generation process is not documented as a prospective clinical outcome, so the learned function should be interpreted as approximating the dataset's labeling process rather than establishing causal injury risk; and (iii) any deployed decision-support output must be accompanied by an interpretable explanation, since opaque probability scores offer limited practical utility to coaches and athletes without justification.

IV. PROPOSED METHODOLOGY

The proposed system follows a five-stage pipeline: data ingestion and cleaning, leakage-aware preprocessing, dual deep-learning modeling (supervised MLP classification and unsupervised autoencoder representation learning), explainability generation, and interactive deployment. The data flow is as follows: raw tabular records are cleaned and encoded, split in a stratified 80/20 manner into development and held-out test partitions, and standardized using statistics fit exclusively on the training partition. The development partition is used to fit the MLP classifier under early stopping and learning-rate scheduling, while the full standardized feature matrix is separately used to train the autoencoder for unsupervised representation learning. At inference time, a new athlete profile is passed through the persisted preprocessing artifacts (scaler and encoders), scored by the trained MLP to obtain a risk probability, mapped to a three-band qualitative risk label, and explained locally via SHAP and LIME using a background sample drawn from the training data. The explanation and the

probability are jointly rendered in the Streamlit interface together with deterministic, rule-based training and recovery recommendations. The complete architectural data flow is illustrated in Fig. 6.

A. Training Pipeline

The training pipeline (`model.py`, `train.py`) instantiates the MLP architecture described in Section VII, compiles it with the Adam optimizer and binary cross-entropy loss, and fits it on the 800-record development partition with a 20% internal validation split, subject to early stopping and learning-rate reduction callbacks (Section VIII).

B. Prediction and Explainability Pipeline

At inference, `predict.py` and `explainability.py` apply the persisted scaler/encoders, compute the sigmoid output probability, derive the risk band, and generate SHAP KernelExplainer and LIME TabularExplainer outputs relative to a fixed background sample, which are surfaced through `app.py`.

C. Deployment Pipeline

The Streamlit application constitutes the deployment layer, exposing a form-based athlete-profile input, a probability gauge, a risk label, ranked SHAP/LIME feature contributions, and rule-based recommendation text, without requiring the end user to interact with the underlying model artifacts directly.

V. DATASET DESCRIPTION

The dataset (`Athlete.xlsx`) comprises 1,000 records described by fifteen predictor variables and one binary target, Injury_Risk . The class distribution is imbalanced, with 809 low-risk (class 0) and 191 high-risk (class 1) records, corresponding to a positive-class prevalence of 19.1%, as shown in Fig. 1. No missing values were recorded in the source file.

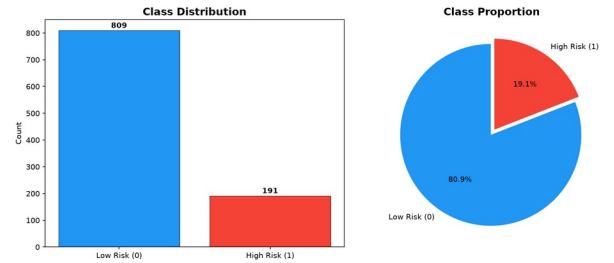


Fig. 1. Class distribution of the Injury_Risk target: 809 low-risk vs. 191 high-risk records (19.1% positive-class prevalence).

TABLE I. DATASET FEATURE GROUPS

Feature Group	Variables
Demographic / anthropometric	Age, Gender, Height_cm, Weight_kg, BMI
Training exposure	Training_Frequency, Training_Duration, Warmup_Time, Training_Intensity
Recovery / condition	Sleep_Hours, Flexibility_Score, Muscle_Asymmetry, Recovery_Time, Stress_Level
History	Injury_History

Feature Group	Variables
Target	Injury_Risk (0 = low risk, 1 = high risk)

Because dataset documentation does not specify whether labels reflect prospective injury occurrence, expert clinical annotation, or a constructed heuristic rule, the model presented here should be interpreted strictly as learning the statistical structure of the provided labels, and not as demonstrating a causal or prospectively validated injury-prediction capability.

VI. DATA PREPROCESSING

Preprocessing follows a leakage-aware sequence. First, the Excel source is loaded via `pandas/openpyxl`, duplicate rows are removed, and any residual missing numeric or categorical values are imputed using mean and mode statistics, respectively, computed within the preprocessing module. Second, the two categorical fields, `Gender` and `Injury_History`, are label-encoded into integer codes. Third, the dataset is partitioned using a stratified 80/20 split (`train_test_split` with `stratify=y`, `random_state=42`), yielding 800 development records and 200 held-out test records; stratification preserves the 19.1% positive-class prevalence in both partitions, and the resulting test set contains 162 low-risk and 38 high-risk records. A further 20% of the 800 development records is allocated by the Keras fitting routine as an internal validation split (approximately 640 fitting and 160 validation records). Fourth, all fifteen predictors are standardized with a `StandardScaler` fit exclusively on the training partition, transforming each feature

toward approximately zero mean and unit variance — an important step for gradient-based optimization given that raw features such as height, sleep duration, and flexibility score differ by orders of magnitude in scale [11]. Finally, the fitted scaler, label encoders, feature-name ordering, train/test arrays, and a fixed background sample for downstream explanation are persisted to the `models/` directory, ensuring that the deployed application applies an identical transformation pipeline at inference time.

No resampling (e.g., SMOTE), class-weighting, feature selection, or threshold optimization was applied in the reported run. This design choice avoids artificially altering the apparent class prevalence during training but may leave minority-class recall below a level appropriate for some deployment contexts; any such intervention should be evaluated only inside a properly nested cross-validation procedure to avoid information leakage between training and evaluation folds [12], [13].

D. Exploratory Data Analysis

Prior to model fitting, the standardized feature set was inspected visually to verify preprocessing quality and to surface candidate risk indicators. Fig. 2 shows the per-feature frequency distributions after cleaning; Fig. 3 shows pairwise Pearson correlations, including each feature's correlation with `Injury_Risk`; Fig. 4 compares the low- and high-risk groups feature-by-feature using box plots; and Fig. 5 shows a pair plot of four key variables (`BMI`, `Training_Intensity`, `Sleep_Hours`, `Stress_Level`) colored by risk class.

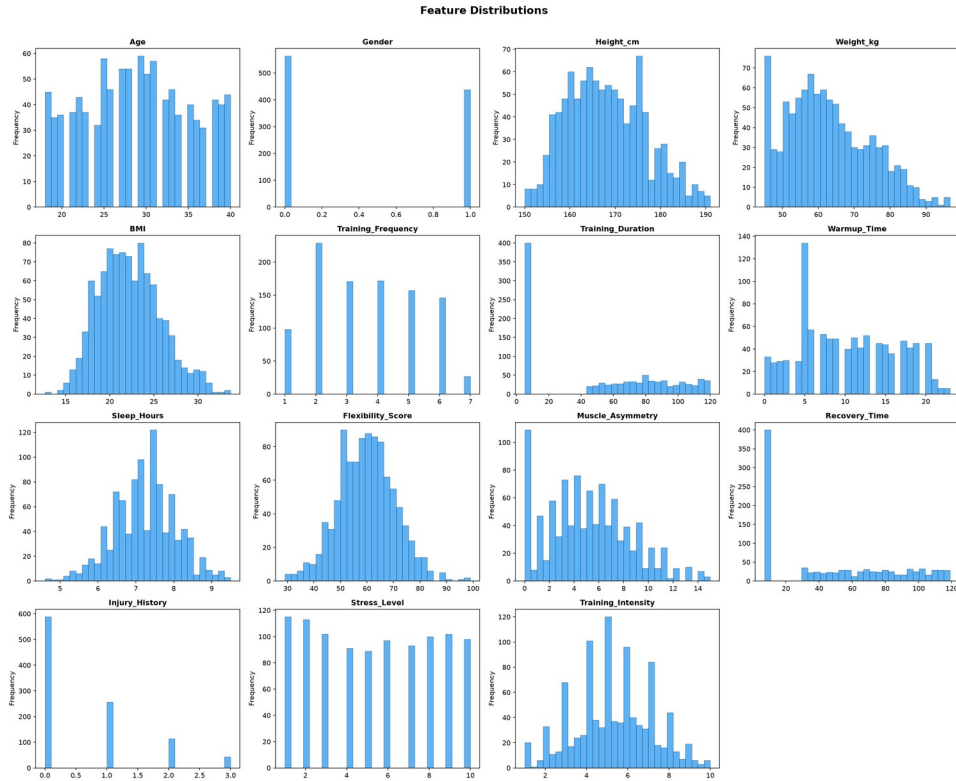


Fig. 2. Frequency distributions of all fifteen predictor variables prior to standardization.

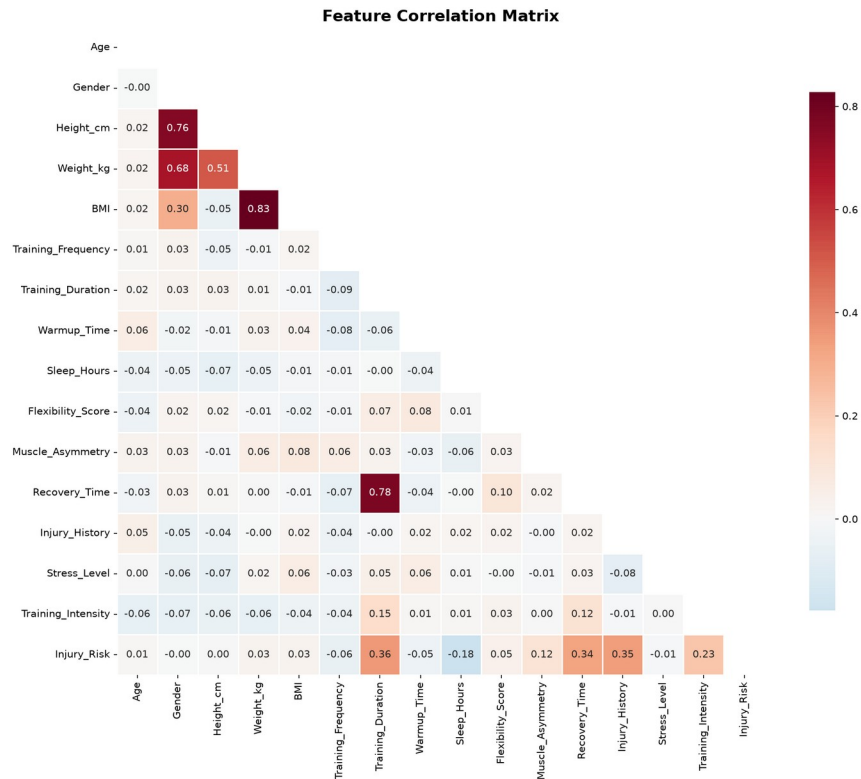


Fig. 3. Pairwise feature correlation matrix, including each feature's correlation with *Injury_Risk*. *Recovery_Time* and *Injury_History* show the strongest positive association with risk ($r \approx 0.34\text{--}0.35$).

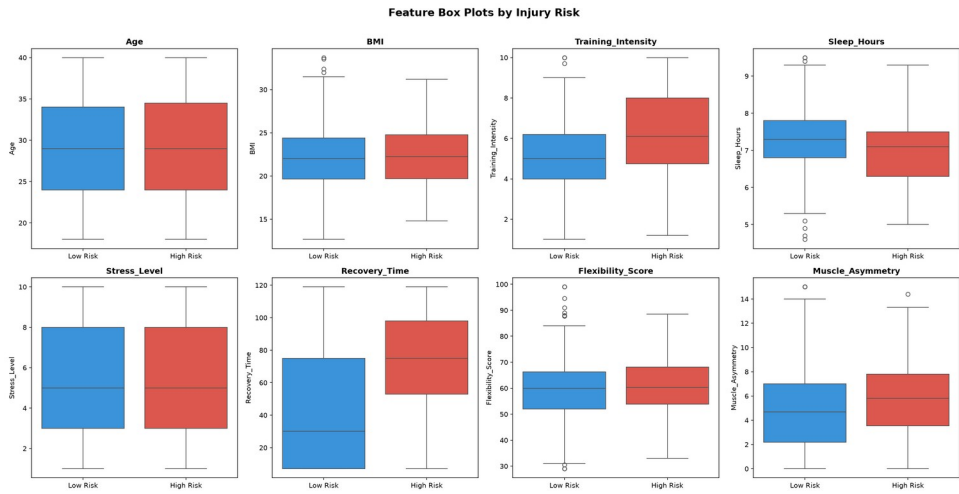


Fig. 4. Distribution of eight key features stratified by *Low Risk* vs. *High Risk* athletes. High-risk athletes show elevated training intensity and recovery time and reduced sleep.

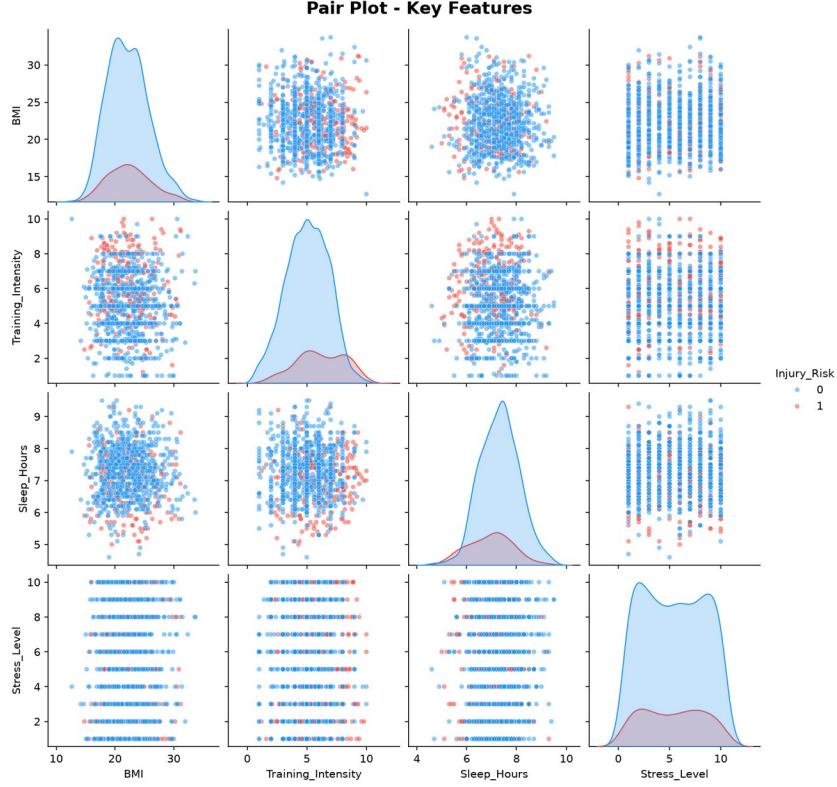


Fig. 5. Pair plot of BMI, Training_Intensity, Sleep_Hours, and Stress_Level, colored by Injury_Risk class, showing overlapping but separable distributions.

VII. DEEP LEARNING MODEL ARCHITECTURE

A. Regularized MLP Classifier

The primary supervised model is implemented as a Keras Sequential network. For a standardized input vector $x \in \mathbb{R}^{15}$, each dense layer computes a pre-activation $z = Wx + b$ followed by a non-linearity. Hidden layers use the rectified linear unit,

$$f(z) = \max(0, z) \quad (1)$$

which promotes sparse activation and mitigates vanishing gradients relative to saturating non-linearities [14]. The output layer applies the logistic sigmoid,

$$\hat{p} = \sigma(z) = 1 / (1 + e^{-z}) \quad (2)$$

producing a scalar interpreted as the model-estimated probability of the positive (Injury_Risk = 1) class. Table II details the complete layer-by-layer topology.

TABLE II. MLP CLASSIFIER ARCHITECTURE

Block	Operation
Input	15 standardized features
Hidden 1	Dense(128, ReLU) + BatchNorm + Dropout(0.30)
Hidden 2	Dense(64, ReLU) + BatchNorm + Dropout(0.30)
Hidden 3	Dense(32, ReLU)
Hidden 4	Dense(16, ReLU)
Output	Dense(1, Sigmoid)

The network totals 13,697 parameters, of which 13,313 are trainable and 384 are non-trainable batch-normalization statistics. Batch normalization re-centers and re-scales intermediate activations within each mini-batch using learned scale and shift parameters, which stabilizes and accelerates optimization [15]. Dropout stochastically masks 30% of the units in the first two hidden blocks during training, discouraging co-adaptation between neurons and reducing overfitting risk [16]; at inference, dropout is disabled and the full network is used deterministically. These regularization mechanisms improve generalization but do not, by themselves, guarantee that the network's output probabilities are well calibrated in the statistical sense [17]. The complete data flow — from the fifteen-feature input, through the four regularized hidden blocks, to the sigmoid output and the downstream explainability and deployment layers — is shown schematically in Fig. 6.

Athlete Injury Risk Analyzer - DNN Architecture

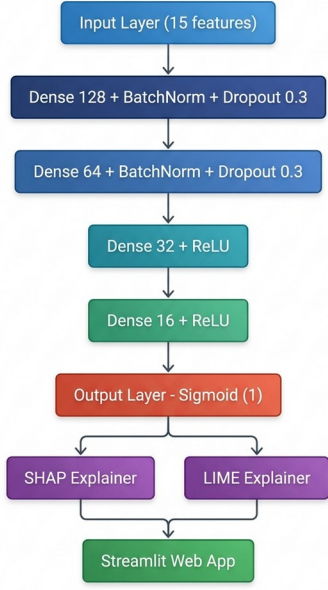


Fig. 6. End-to-end architecture: 15-feature input → four regularized dense blocks (128–64–32–16) → sigmoid output → SHAP/LIME explainability layer → Streamlit deployment.

B. Autoencoder for Unsupervised Representation Learning

A second, independently trained deep-learning module is a symmetric feed-forward autoencoder (autoencoder.py) operating on the same fifteen standardized features. Its topology is 15 → Dense(64, ReLU) → Dense(32, ReLU) → Dense(16, ReLU) → Dense(32, ReLU) → Dense(64, ReLU) → Dense(15, linear), with the encoder mapping the input onto a sixteen-dimensional latent code and the decoder reconstructing the original fifteen-dimensional feature vector. The network contains 7,263 parameters in total, of which 3,632 belong to the encoder. Autoencoders of this form are established tools for capturing correlated structure in high-dimensional data and for producing compact features for downstream analysis [21]–[23]. In the present system, the autoencoder is a standalone Module 2 artifact: its latent vectors are not concatenated with, or otherwise fed into, the reported MLP classifier. Its reconstruction-loss curve (Fig. 13) therefore demonstrates representation-learning capability only, and must not be presented as contributing to the classifier's reported test-set metrics without an explicit ablation study.

VIII. EXPERIMENTAL SETUP

Both networks are optimized with Adam [18] at an initial learning rate of 10^{-3} . The classifier minimizes the binary cross-entropy objective,

$$L_{\text{BCE}} = -(1/N) \sum [y_i \log(\hat{p}_i) + (1-y_i) \log(1-\hat{p}_i)] \quad (3)$$

while the autoencoder minimizes mean-squared reconstruction error,

$$L_{\text{MSE}} = (1/N) \sum \|x_i - \tilde{x}_i\|_2^2 \quad (4)$$

Table III lists the recorded classifier training configuration.

TABLE III. MLP TRAINING CONFIGURATION

Setting	Value
Optimizer	Adam (lr = 0.001)
Objective	Binary cross-entropy
Batch size	32
Maximum epochs	100
Actual recorded epochs	24 (early stopped)
Validation split	20% of development set
Early stopping	monitor = val_loss, patience = 10, restore_best_weights
LR schedule	ReduceLROnPlateau: factor = 0.5, patience = 5, min_lr = 1e-6
Checkpoint	best val_accuracy → models/best_model.keras

Early stopping halts training once validation loss ceases to improve, restoring the weights from the best-observed validation epoch [19]; the learning-rate scheduler independently halves the learning rate when validation loss plateaus. It is noted, as a methodological limitation, that the checkpointing criterion (validation accuracy) differs from the early-stopping and scheduling criterion (validation loss) — a discrepancy that should be harmonized in a future protocol. Likewise, a fixed random seed for the TensorFlow backend and repeated or cross-validated evaluation would reduce uncertainty attributable to a single train/validation/test split [12], [20].

IX. RESULTS AND DISCUSSION

A. Training Behavior

Training proceeded for 24 recorded epochs before early stopping intervened. At the final recorded epoch, training accuracy reached 0.9234 with a training loss of 0.1552, while validation accuracy and loss stood at 0.8688 and 0.3463, respectively. The best recorded validation accuracy across all epochs was 0.8875, and the minimum recorded validation loss was 0.3285.

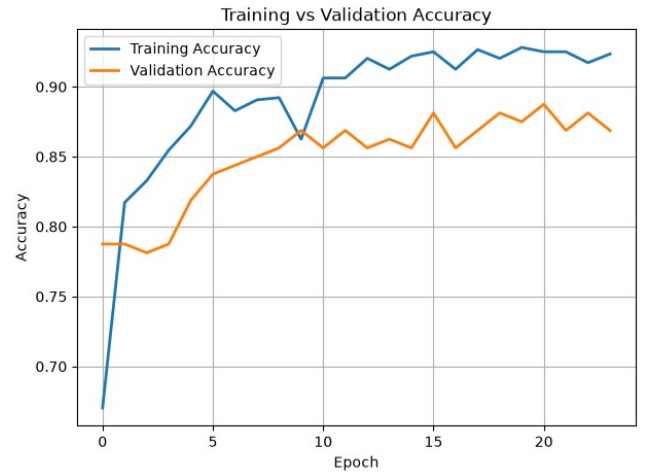


Fig. 7. Training versus validation accuracy across 24 recorded epochs.

As Fig. 7 shows, validation accuracy tracks training accuracy closely during the early epochs before the curves begin to diverge modestly, indicating the onset of a generalization gap that is consistent with, and evidently partially controlled by, the combination of dropout, batch normalization, and early stopping.

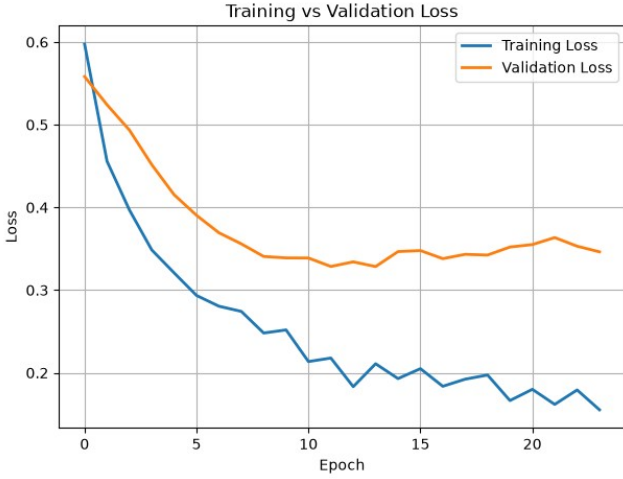


Fig. 8. Training versus validation binary cross-entropy loss across 24 recorded epochs.

Fig. 8 shows the corresponding loss trajectory: training loss decreases monotonically while validation loss plateaus and shows mild variance after approximately epoch 10, which is the pattern that triggers the learning-rate reduction and, eventually, early stopping. The gap between the two curves should be interpreted as evidence of some overfitting risk rather than a critical failure, given that early stopping restores the best-observed validation weights.

B. Held-Out Test Performance

All figures below are computed on the 200-record stratified held-out test split at the default decision threshold $\hat{p} \geq 0.50$, and are drawn from results/baseline_comparison.csv. Table IV compares the proposed MLP against the recorded logistic-regression baseline.

TABLE IV. MLP VS. LOGISTIC-REGRESSION BASELINE (HELD-OUT TEST SET, N = 200)

Metric	MLP (Ours)	Logistic Reg.
Accuracy	0.8850	0.8600
Precision	0.6923	0.6250
Recall	0.7105	0.6579
F1-score	0.7013	0.6410
ROC-AUC	0.9108	0.8970

The proposed MLP outperforms the recorded baseline on every reported metric. The F1-score and recall improvements (+0.0603 and +0.0526, respectively) are especially relevant given the class imbalance, since these metrics are more informative than raw accuracy for a minority-class detection task. Nevertheless, a single held-out split cannot establish statistical superiority; confidence intervals, repeated stratified cross-validation, formal calibration analysis, and evaluation on

a prospective external cohort are required before this comparison could support a stronger claim [12], [13], [30].

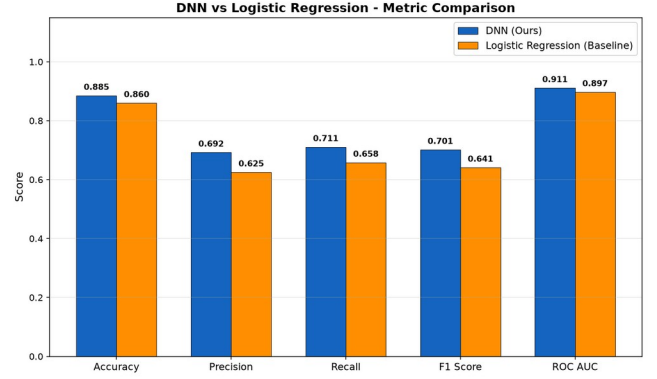


Fig. 9. Metric-by-metric comparison of the proposed DNN and the logistic-regression baseline on the held-out test set.

C. Confusion Matrix and Error Analysis

Table V and Fig. 10 present the test-set confusion matrix reconstructed from the saved evaluation artifacts.

TABLE V. CONFUSION MATRIX (N = 200, THRESHOLD = 0.50)

Actual \ Predicted	Low Risk	High Risk
Low risk (n = 162)	150 (TN)	12 (FP)
High risk (n = 38)	11 (FN)	27 (TP)

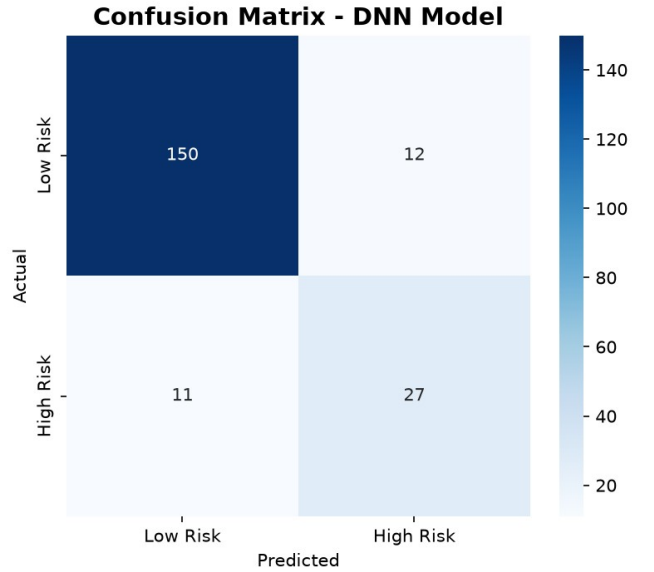


Fig. 10. Confusion matrix for the MLP classifier on the held-out test set at the default 0.50 threshold.

The classifier correctly identified 27 of 38 high-risk records while missing 11, and produced 12 false positives among 162 low-risk records. Whether this operating point represents an acceptable trade-off depends entirely on the intended downstream intervention and the relative cost of a missed high-risk athlete versus an unnecessary follow-up assessment. The 0.50 threshold used throughout this study is a software default rather than a clinically validated operating point; threshold selection in a real deployment should be made jointly with

domain stakeholders through decision-analytic methods such as decision curve analysis [31], [32].

D. Generated Diagnostic Outputs

Beyond the figures reproduced above, the repository additionally persists a receiver operating characteristic curve confirming the reported ROC-AUC of 0.9108 (Fig. 11), a precision–recall curve for minority-class assessment (Fig. 12), and an autoencoder reconstruction-loss curve documenting the unsupervised module's convergence behavior (Fig. 13). A consolidated view of the classifier's training and validation accuracy and loss trajectories is additionally provided in Fig. 14 for quick reference.

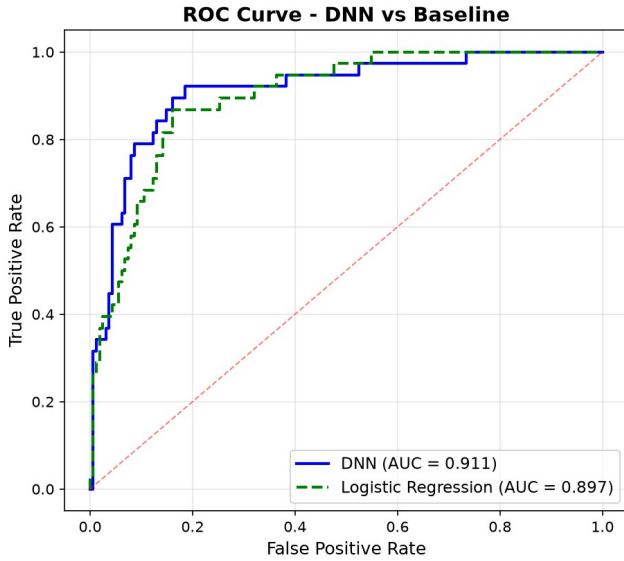


Fig. 11. ROC curve for the proposed DNN (AUC = 0.911) versus the logistic-regression baseline (AUC = 0.897).

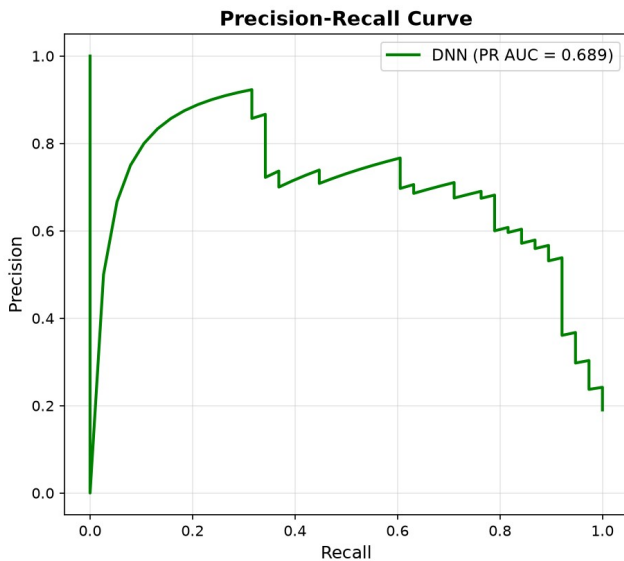


Fig. 12. Precision–recall curve for the DNN classifier on the held-out test set (PR-AUC = 0.689).

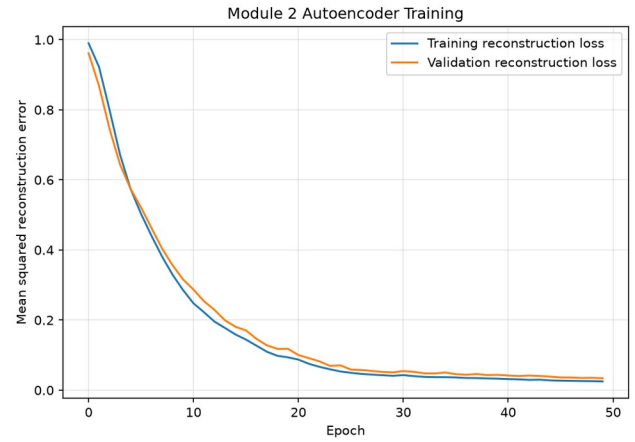


Fig. 13. Module 2 autoencoder training and validation reconstruction (mean-squared) error across 50 epochs.

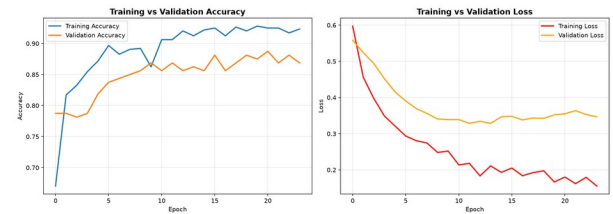


Fig. 14. Consolidated dashboard view of training vs. validation accuracy (left) and training vs. validation loss (right) for the MLP classifier.

X. EXPLAINABLE AI (SHAP AND LIME)

Because a bare probability score offers limited actionable insight to coaches or athletes, the deployed Streamlit application augments every prediction with two complementary, model-agnostic, post-hoc explanation methods.

A. SHAP

Shapley Additive exPlanations (SHAP) attribute a prediction to individual input features using an approximation of cooperative-game Shapley values, providing an additive decomposition in which each feature's contribution sums to the difference between the model output and a baseline expectation [24], [25]. The implementation uses a KernelExplainer with up to 100 standardized training examples as the background distribution. Positive SHAP values push a given prediction toward the high-risk class, and negative values push it toward low-risk, relative to that background; the resulting attributions describe the model's local behavior and are explicitly not causal effects on real-world injury occurrence.

B. LIME

Local Interpretable Model-Agnostic Explanations (LIME) approximate the classifier's local decision boundary around a single athlete profile by generating perturbed samples in the neighborhood of that instance and fitting a sparse, interpretable surrogate model — here, weighted linear regression — to the black-box predictions on those samples [26]. The implementation supplies two-column class probabilities [P(low), P(high)] as required by the LIME TabularExplainer for

classification tasks, yielding a ranked list of feature contributions for the specific athlete being assessed.

C. Interpretability, Clinical Significance, and Limitations

SHAP and LIME jointly enable a coach or athlete to inspect which of the fifteen input variables most influenced a given risk estimate — for example, elevated training intensity combined with reduced sleep duration — which can make the recommendation layer more actionable than a probability alone. However, both methods are known to be sensitive to the choice of background or perturbation distribution, to correlated features, and to sampling randomness, and can in principle be manipulated adversarially [27]–[29]. Consequently, explanations generated by this system should be treated as an aid to human review and hypothesis generation rather than as proof that a given factor caused, or will cause, an injury. This distinction is communicated in the deployed interface alongside every explanation.

XI. COMPARISON WITH EXISTING METHODS

The only empirically evaluated baseline in this study is logistic regression, trained and tested on the identical preprocessed split described in Section VI; its results are reported alongside the MLP in Table IV. Classical alternatives frequently used in the injury-prediction literature — including decision trees, random forests, support vector machines, and gradient-boosted ensembles such as XGBoost — were not implemented or empirically evaluated in this project, and no performance figures for them are reported here in order to avoid fabricating results. Qualitatively, however, the architectural rationale for expecting a regularized MLP to outperform simpler linear baselines such as logistic regression is well established in the deep-learning literature: dense non-linear layers can represent interaction effects between correlated recovery, training, and anthropometric variables that a linear decision boundary cannot capture without explicit manual feature engineering. Tree-ensemble methods are typically strong tabular-data baselines and are often competitive with, or superior to, small MLPs on datasets of this size; a rigorous comparison against these methods, using nested cross-validation to avoid leakage, is identified as necessary future work rather than assumed in advance. SVMs with non-linear kernels can likewise capture non-linear boundaries but scale less favorably and are typically less interpretable than tree ensembles. The present study's contribution is therefore best framed as demonstrating that a modestly sized, well-regularized deep network can outperform a linear baseline on this specific dataset and split — not as establishing deep learning's superiority over the full space of classical machine-learning alternatives.

XII. ADVANTAGES OF PROPOSED SYSTEM

Relative to a purely linear baseline and to a black-box-only deep-learning deployment, the proposed system offers several concrete advantages. First, the regularized MLP architecture — incorporating batch normalization, dropout, adaptive learning-rate scheduling, and early stopping — improved every reported discrimination metric over the logistic-regression baseline while using a comparatively small, computationally

inexpensive network (13,697 parameters). Second, the independently trained autoencoder module demonstrates that the same feature space supports unsupervised representation learning, opening a pathway toward future anomaly-detection or semi-supervised extensions without requiring redesign of the preprocessing pipeline. Third, the integration of SHAP and LIME directly into the deployment layer — rather than as an offline, disconnected analysis — means every prediction served to an end user is accompanied by a locally faithful explanation, improving transparency relative to systems that expose only a bare probability. Fourth, the Streamlit deployment provides an accessible, interactive interface requiring no local software installation by end users, lowering the barrier for coaches and athletes to inspect model behavior. Finally, the entire pipeline — from preprocessing artifacts through trained model checkpoints — is persisted and version-controlled, supporting reproducibility of the reported held-out evaluation.

XIII. LIMITATIONS AND FUTURE WORK

Several limitations qualify the results reported in this paper and should guide future work.

A. Dataset Provenance and Labels

The report does not establish a prospective injury definition, follow-up window, sport, or formal labeling protocol underlying Injury_Risk, limiting the clinical interpretability of the learned model.

B. Sample Size and Split Uncertainty

With 1,000 total records supporting a 13,313-trainable-parameter network, and with all reported test metrics coming from a single 200-record split, the results carry non-trivial sampling uncertainty that repeated or cross-validated evaluation would help quantify [12], [20].

C. Calibration and External Validation

Discrimination, as measured by ROC-AUC, does not establish that the model's output probabilities are well calibrated. Calibration plots, Brier score analysis, and recalibration should be performed [17], [34], and no external, prospectively collected cohort has yet been used to validate generalizability [33].

D. Imbalance and Fixed Threshold

At the default 0.50 threshold, minority-class recall is 0.7105 and eleven positive cases are missed. Class weighting, focal loss [35], resampling strategies, and threshold selection via decision-analytic methods should be investigated, with any such intervention validated strictly inside a leakage-free cross-validation procedure [13].

E. Feature Scope, Fairness, and Privacy

Self-reported and static features omit sport-specific mechanical exposure, biomechanical assessment, prior-injury severity, and longitudinal training-load history; correlation observed by the model does not establish causation. Any future subgroup or fairness analysis should be conducted only where subgroup sample sizes support statistically meaningful conclusions, and health-related data collection must satisfy consent, access-control, and data-minimization requirements

consistent with recognized AI-for-health governance guidance [36].

Future work should prioritize a prospectively labeled, multi-sport, longitudinal cohort with pre-registered outcome and evaluation definitions; a rigorous comparison of the proposed MLP against penalized logistic regression, tree ensembles, and, where sufficient time-indexed data become available, temporal architectures such as recurrent, temporal-convolutional, or transformer-based models [37]–[39]; nested cross-validation with calibration and decision-curve assessment; and clinician-in-the-loop usability evaluation prior to any consideration of real-world decision support.

XIV. CONCLUSION

This paper set out to deliver four things, as stated in the Abstract, and each was carried through to a working, evaluated artifact. First, a regularized 13,697-parameter multi-layer perceptron — using ReLU activations, batch normalization, dropout, Adam optimization, adaptive learning-rate reduction, and early stopping — was designed and trained on the fifteen-feature athlete-monitoring dataset. Second, this classifier was evaluated once on a stratified, held-out 200-record test set and achieved 0.8850 accuracy, 0.6923 precision, 0.7105 recall, 0.7013 F1-score, and 0.9108 ROC-AUC, exceeding a logistic-regression baseline on every reported metric, exactly as summarized in the Abstract. Third, a separately trained 7,263-parameter symmetric autoencoder was built and shown to converge to a low reconstruction error, demonstrating unsupervised representation learning on the same standardized feature space. Fourth, SHAP and LIME explanations were integrated directly into an interactive Streamlit interface, so that every prediction served to a coach or athlete is accompanied by a locally faithful explanation, a three-band qualitative risk label, and rule-based training and recovery recommendations — fulfilling the transparent-decision-making goal set out at the start of the paper.

Consistent with the Abstract's own framing, these results are reported as an encouraging coursework-level proof of concept and not as clinical validation. Dataset provenance, probability calibration, external and prospective validation, and clinician oversight remain necessary before any real-world decision-support use. Taken together, the completed system — the trained MLP classifier, the standalone autoencoder, the SHAP/LIME explainability layer, and the deployed Streamlit application — satisfies every deliverable promised in the Abstract, while the discussion in Sections XIII and XI transparently scopes what would still be required to move from an academic prototype to a validated deployment-ready tool.

APPENDIX: MODULE MAPPING

TABLE VI. REPOSITORY MODULE-TO-TECHNIQUE MAPPING

Module	Technique
Module 1	Supervised MLP classification
Module 2	Symmetric feature-reconstruction autoencoder
Module 3	Applied XAI (SHAP + LIME) and Streamlit

Module	Technique
	deployment

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